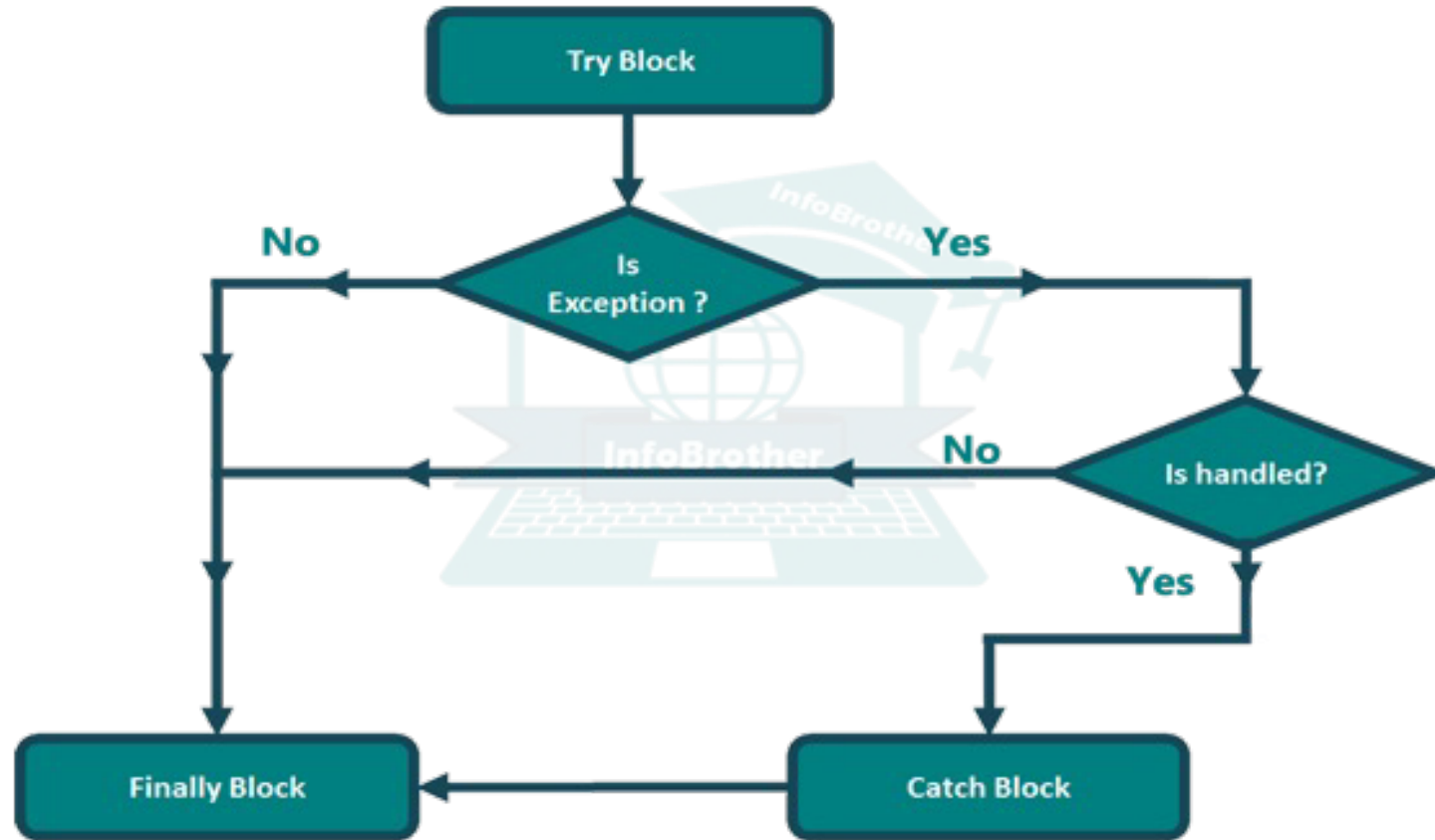


Exceptions

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Exception Handling



Reference: https://www.infobrother.com/Tutorial/C++/C++_Exception_Handling

Uncaught exception

```
1  #include<iostream>
2  #include<stdexcept>
3  #include <vector>
4  using namespace std;
5
6  int main() {
7      std::vector<int> myvector(10);
8      int n = myvector.at(20);
9  }
```

```
lincyu@Chih-YudeMacBook-Pro Chapter14 % ./a.out
libc++abi: terminating with uncaught exception of type std::out_of_range:
vector
zsh: abort      ./a.out
```

try / catch

```
1  #include<iostream>
2  #include<stdexcept>
3  #include <vector>
4  using namespace std;
5
6  int main() {
7      std::vector<int> myvector(10);
8      try {
9          int n = myvector.at(20);
10         cout << "OK!" << endl;
11     } catch (const out_of_range &e) {
12         cerr << e.what() << endl;
13     }
14 }
```

Reference

- stdexcept
 - <https://cplusplus.com/reference/stdexcept/>
- Exception Handling
 - https://www.infobrother.com/Tutorial/C++/C++_Exception_Handling

throw

```
4  int main()
5  {
6      /*Enter the value of x: if x>0, program execute normally:
7       if x<0: the program will catch the error and display the error:*/
8      int x;
9      cout<<" Enter the value of x: ";
10     cin>>x;
11     cout<<" Before try"<<endl;
12     try {
13         cout<<" Inside try"<<endl;
14         if(x < 0) { //check the code here:
15             throw x; //if x<0, throw x:
16             //after throw statement: nothing will execute in try block ever:
17             cout<<"After throw"<<endl; //will not execute ever:
18         }
19     } catch (int x) { //if try throw any exception, catch it:
20         //if the program will catch exception: then the code inside this block will execute:
21         cout<<" Exception Caught: " << x <<endl;
22         //handle the exception here,
23     }
24
25     cout<<" After catch" <<endl;
26     return 0;
27 }
```

More example

```
4  double fact(int n) {
5      if (n > 170 || n < 0) {
6          throw n;
7      }
8      double total = 1;
9      for (int i = 1; i <= n; i++) {
10         total *= i;
11     }
12     return total;
13 }
14
15 int main() {
16     int x = 5, y = -2;
17
18     try {
19         cout << "C(" << x << ", " << y << ") = "
20         << fact(x)/(fact(x-y)*fact(y)) << endl;
21     } catch (int i) {
22         cout << "Exception: " << i << endl;
23     }
24 }
```

Customized Exception

```
4  class BufferFull: public exception {};  
5  
6  class IntBuffer {  
7  private:  
8      int buffer[2];  
9  public:  
10     int count;  
11     void add(int value) {  
12         if (count == 2) {  
13             throw BufferFull();  
14         }  
15         buffer[count++] = value;  
16     }  
17 };
```

```
19 int main() {  
20     IntBuffer buf;  
21     buf.count = 0;  
22     try {  
23         buf.add(1);  
24         buf.add(2);  
25         // buf.add(3);  
26     } catch (BufferFull e) {  
27         cout << "BufferFull" << endl;  
28     }  
29 }
```